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Statement of Purpose

The recent success of large language models (LLMs) is attributed to their training on extensive data. While LLMs are voraciously learning existing knowledge through this paradigm, I believe true intelligence lies more in creating discoveries, methods, and theories that push the boundaries of the sum of all human knowledge. Despite studies showing LLMs can be guided to create certain new knowledge when subtly prompted, whether LLMs can do so autonomously and systematically remains unclear. How can LLMs move beyond *learning existing knowledge from humans* to *creating new knowledge for humans*? This motivates my long-term ambition of exploring a new paradigm of ***self-evolving* LLMs**, with a specific focus on developing **LLMs capable of scientific research** as its practical context. Its rationale stems from how human researchers create new knowledge and improve themselves simultaneously. Research exploration not only leads to productivity but also brings up research experience, through which we develop our understanding and skills continuously. Drawing a parallel, the self-evolving LLM explores new knowledge, while simultaneously improving its knowledge and capabilities through its firsthand research experience.

The paradigm of self-evolving LLMs addresses two challenges unresolved by merely scaling up LLM training within the current paradigm. First, creating new knowledge is a path unknown to humans — humans' existing knowledge does not necessarily lead to specific new knowledge or guide us in achieving it. Thus, **inherently limited is the supervision** provided by data derived from humans' existing knowledge, potentially confining LLMs within boundaries yet to be transcended by humans. The self-evolving LLM could overcome the challenge as it navigates and creates its own path toward new knowledge, drawing on its own research experience as a source of learning. Second, even if the supervision does not confine LLMs, a practical challenge still remains. We are **running out of training data** as we continue to scale up its size [1], meaning it will soon be impractical to heavily rely on available data. In contrast, self-evolving should be agnostic to data exhaustion, as the LLM's firsthand research experience serves as the dynamically acquired data.

Framework I envision a self-evolving LLM that: (1) dynamically acquires **research experience as data**; (2) conducts **self-guided training**, updating its parameters by acquired experience, with a **modularized architecture** to overcome catastrophic forgetting; and (3) is **reliably evaluated**.

Research Experience as Data Human research experience has two sources: (1) our internal *thinking* processes guiding the research, including activities from setting goals, through brainstorming ideas and thought experiments, and even down to writing code for a specific experiment; and (2) our *interaction* with the external world, including conducting experiments and communicating with collaborators and peers. Thinking involves planning the interaction and interaction provides external feedback to thinking. Similarly, the self-evolving LLM could also dynamically acquire research experience through continuous cycles of thinking and interaction.

I will explore various specific sources of research experience for LLMs. For example, much of a human researcher's internal thinking revolves around consistent introspection. I always introspectively assess the novelty or validity of an idea before executing it. While most introspective activities do not lead to groundbreaking insights, they offer invaluable research experience to me. However, existing LLMs struggle with some low-level introspective activities [2], making introspection, a vital source of research experience, currently unfeasible. One of my subgoals is thus to improve that.

Self-Guided Training and Modularized Architecture We need a training-based framework to accumulatively convert all research experience into learned experiential knowledge, as merely augmenting the input context does not internalize what the LLM learned, especially given the inherent constraints on context length. I will explore "how to train" and "what architecture to use".

How do we convert the dynamically acquired research experience into text data points for LLM training, and what training algorithms do we use? My ultimate goal is self-guided training. Specifically, the LLM autonomously adapts its training for diverse and dynamically changing research scenarios, without being limited by potentially sub-optimal predefined protocols. For example, the LLM could decide to convert certain pieces of experience into data points with the structure of input, output, and an additional natural language explanation for that output. Each data point's explanation bridges its input and output explicitly, while collectively, these explanations can be transformed into a broader understanding for the LLM to learn. Accordingly, the LLM could decide to adopt the training algorithm from my paper [3], which is designed to enable this transformation. Self-understanding is indispensable to self-guided training. It allows the LLM to frame each research experience within the context of its current capabilities, working mechanisms, and characteristics,

empowering the LLM to dynamically optimize the training process tailored to itself. This motivates my subgoals: studying how LLMs understand themselves, how to enhance their self-understanding, and how this can improve our humans' understanding.

As the LLM continuously updates its parameters, using data whose distribution differs significantly from its prior training, it will suffer from catastrophic forgetting. I want to explore the modularized architecture of LLMs as a potential solution, where each architectural module is responsible for specific capabilities. As each research experience enhances very specific LLM capabilities, such as understanding Python running logs and searching for relevant literature, we could update only relevant modules in the corresponding iteration. There would be a core module responsible for essential functions, like deciding which modules to update. The overall idea is inspired by my paper [4], which showed pretrained transformers naturally manifest emergent modularity without deliberate design, and distinct capabilities are anchored to specific modules. My paper [5] also designed a lightweight pluggable module (a small linear layer) that effectively enhances pretrained LMs with fresh factual knowledge, even though the pretrained LM and the module have never been trained to cooperate with each other. They both suggest the feasibility of a post-pretraining modularization, circumventing inductive biases that might hamper scaling up the LLM.

Evaluation Evaluation is always crucial but challenging. For example, many researchers are using LLMs as automatic evaluators to compare two models' responses. However, when asked to choose between two responses, one adhering to a given instruction and the other deviating from it, most LLMs, even with advanced prompting strategies, are significantly outperformed by random guessing on a carefully designed benchmark introduced by my paper [6]. This is because LLMs tend to be cheated by such responses: they are deliberately designed to look appealing yet actually deviate from the instruction, causing the LLMs to fail to detect this deviation. It then raises doubt about the reliability of many mainstream LLM leaderboards that rely on LLM evaluators.

In my pursuit of developing self-evolving LLMs, evaluation poses even greater challenges, because evaluating whether LLMs achieve my goal and many of my subgoals goes beyond human (or an average person's) capabilities. For example, assessing the introspection of LLMs is challenging. Using static datasets to measure "correctness" conflicts with the goal of creating new knowledge. Moreover, a person would require extensive domain knowledge beyond this of the LLM to directly assess the LLM's introspection. To this end, I will explore crafting indirect tasks that are more evaluable, and observing the performance of the LLM or LLM-assisted humans. For example, when evaluating the LLM's self-understanding, we humans cannot understand the workings of black-box LLMs. A possible indirect task might involve predicting the LLM's behavior after updating a particular module, and then seeing how the performance of humans (assisted by the LLM itself) reveals insights into the LLM's self-understanding.

Why UW Aiming to become a professor at a top university, I regard UW, with its leading NLP community of talented faculty and students, as the perfect place for my Ph.D. Moreover, I greatly appreciate the collaborative and inclusive atmosphere of UW's NLP community, which I believe will broaden my research perspective and empower me to conduct impactful research.

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